

1. Instrumental Variable

a. A valid instrument if satisfied three main conditions

- i. Relevance: $\text{Cov}(Z, A) \neq 0$
- ii. Exclusion Restriction: $Y(a, z) = Y(a, z^*) = Y(a) \forall z \text{ and } z^* \forall a$
- iii. Independence: $Z \perp U$

b. The assumption (A1) – (A3) are insufficient for identification.

- i. Effect Homogeneity: $Y(A = 1) - Y(A = 0) \perp Z$
- ii. Monotonicity: $A(Z = 1) \geq A(Z = 0)$ for all individuals (No Defiers)

2. From the definition, the model is said to be overidentified if

No. of Instrument = $m > k$ = No. of endogenous regressor

a. $m = 3 > 1 = k$. It meets overidentified case.

b. The null and alternative hypotheses for Sargan Test are

$$H_0: E(\mathbf{Z}^\top \boldsymbol{\varepsilon}) = 0 \text{ vs } H_1: E(\mathbf{Z}^\top \boldsymbol{\varepsilon}) \neq 0$$

and the Test statistic is

$$nR^2 \sim \chi^2_{(m-k)}$$

where the R-squared R^2 is obtained by regressing $\hat{\boldsymbol{\varepsilon}}$ against Z_1, Z_2, Z_3 .

c. At the 5% significance level

$$nR^2 = 500 \times 0.012 = 6 > 5.99 = \chi^2_{(3-1), 0.95}$$

$$\Rightarrow \text{reject } H_0$$

At least one of instruments is invalid.

3. Based on the approximation identity

$$\frac{\text{Var}(\hat{\beta}_{IV})}{\text{Var}(\hat{\beta}_{OLS})} \approx \frac{1}{R^2_{Z \sim A}}$$

a. When $R^2_{Z \sim A}$ is large, the instrument is strong. Strong instruments lead to narrower interval estimates.

b. From the identity and $\text{se}(\hat{\beta}_{IV}) = 2$

$$\frac{\text{se}(\hat{\beta}_{IV})}{\text{se}(\hat{\beta}_{OLS})} = \frac{2}{\text{se}(\hat{\beta}_{OLS})} = \sqrt{\frac{1}{0.01}} = 10$$

$$\Rightarrow \text{se}(\hat{\beta}_{OLS}) = 0.2$$

c. For IV $\Rightarrow 7 \pm (2 \times 2) = [3, 11]$

For OLS $\Rightarrow 7 \pm (2 \times 0.2) = [6.6, 7.4]$

4. Based on regression approach

a. Difference Method

$$\text{TE} = 1.2, \quad \text{NIE} = 1.2 - 0.7 = 0.5$$

b. The Proportion Mediated is

$$\text{PM} = \frac{\text{TE} - \text{NDE}}{\text{TE}} = \frac{0.5}{1.2} < 0.5$$

The NDE is dominant because its contribution is greater than indirect pathway.

5. At 5% significance level

a. For α , the t-statistic is

$$T_\alpha = \frac{0.45}{0.15} = 3$$

For β , the t-statistic is

$$T_\beta = \frac{0.2}{0.12} = \frac{5}{3}$$

b. Suppose H_0 is expressed as a union

$$H_0: \theta \in \bigcup_{d \in D} \Theta_d$$

Suppose that for each $d \in D$, $\{\mathbf{X} | T_d(\mathbf{X}) \in R_d\}$ is rejection region for a test of

$$H_{0d}: \theta \in \Theta_d \text{ vs } H_{1d}: \theta \in \Theta_d^c$$

Then the rejection region for IUT of H_0 is

$$\bigcap_{d \in D} \{\mathbf{X} | T_d(\mathbf{X}) \in R_d\}$$

c. Joint significant test (IUT)

$$T_\alpha \wedge T_\beta = \frac{5}{3} < 1.96$$

It does not reject H_0 , the indirect effect is zero.

6. A series of assumptions are crucial to avoid confounding

a. Exchangeability

- i. No unmeasured $A - Y$ confounding. $Y(a, m) \perp A | C$
- ii. No unmeasured $A - M$ confounding. $M(a) \perp A | C$
- iii. No unmeasured $M - Y$ confounding. $Y(a, m) \perp M | A, C$
- iv. Cross-world exchangeability. $Y(a, m) \perp M(a^*) | C$

b. The Cross-world exchangeability failed when there exist $M - Y$ confounder L affected by exposure A . Consider NPSEM as follows

$$A = g_A(\varepsilon_A), \quad L = g_L(A, \varepsilon_L), \quad M = g_M(A, L, \varepsilon_M), \quad Y = g_Y(A, L, M, \varepsilon_Y)$$

For $Y(a, m)$, let $A = a$, $M = m$

$$L(a) = g_L(a, \varepsilon_L)$$

$$Y(a, m) = g_Y(a, L(a), m, \varepsilon_Y) = g_Y(a, g_L(a, \varepsilon_L), m, \varepsilon_Y) = f_Y(\varepsilon_L, \varepsilon_Y)$$

For $M(a^*)$, let $A = a$

$$L(a^*) = g_L(a^*, \varepsilon_L)$$

$$M(a^*) = g_M(a, L(a^*), \varepsilon_M) = g_M(a, g_L(a^*, \varepsilon_L), \varepsilon_M) = f_M(\varepsilon_L, \varepsilon_M)$$

Both $Y(a, m)$ and $M(a^*)$ depend on the same shared error term ε_L . Because they share a common random component, they are correlated. Therefore, the assumption $Y(a, m) \perp M(a^*)$ fails.

7. The mediation formula is

$$Q(a, a^*) = \sum_c \left\{ \int_m E(Y|M = m, A = a, C = c) f_{M|A,C}(m|a^*, c) dm \right\} P(C = c)$$

Given that

$$\begin{aligned} \log\{E(Y|M = m, A = a, C = c)\} &= \beta_0 + \beta_1 a + \beta_2 m + \beta_c c \\ M|A = a^*, C = c &\sim N(\alpha_0 + \alpha_1 a^* + \alpha_c c, 2[1 + a^*]\sigma^2) \end{aligned}$$

a. The exact form for $Q(a, a^*)$

$$\begin{aligned} Q(a, a^*) &= \sum_c \left\{ \int_m e^{\beta_0 + \beta_1 a + \beta_2 m + \beta_c c} f_{M|A,C}(m|a^*, c) dm \right\} P(C = c) \\ &= e^{\beta_0 + \beta_1 a} \sum_c e^{\beta_c c} \left\{ \int_m e^{\beta_2 m} f_{M|A,C}(m|a^*, c) dm \right\} P(C = c) \\ &= e^{\beta_0 + \beta_1 a} \sum_c e^{\beta_c c} E(e^{\beta_2 M} | A = a^*, C = c) P(C = c) \\ &= e^{\beta_0 + \beta_1 a} \sum_c e^{\beta_c c} (e^{\beta_2(\alpha_0 + \alpha_1 a^* + \alpha_c c) + \beta_2^2 \sigma^2 (1 + a^*)}) P(C = c) \\ &= e^{\beta_0 + \beta_1 a + \beta_2(\alpha_0 + \alpha_1 a^*) + \beta_2^2 \sigma^2 (1 + a^*)} \sum_c e^{(\beta_c + \beta_2 \alpha_c) c} P(C = c) \\ &= e^{\beta_0 + \beta_1 a + \beta_2(\alpha_0 + \alpha_1 a^*) + \beta_2^2 \sigma^2 (1 + a^*)} E(e^{(\beta_c + \beta_2 \alpha_c) C}) \end{aligned}$$

Let $\Lambda = E(e^{(\beta_c + \beta_2 \alpha_c) C})$ then

$$\begin{aligned} Q(1,1) &= \Lambda e^{\beta_0 + \beta_1 + \beta_2(\alpha_0 + \alpha_1) + 2\beta_2^2 \sigma^2} \\ Q(1,0) &= \Lambda e^{\beta_0 + \beta_1 + \beta_2 \alpha_0 + \beta_2^2 \sigma^2} \\ Q(0,1) &= \Lambda e^{\beta_0 + \beta_2(\alpha_0 + \alpha_1) + 2\beta_2^2 \sigma^2} \\ Q(0,0) &= \Lambda e^{\beta_0 + \beta_2 \alpha_0 + \beta_2^2 \sigma^2} \end{aligned}$$

b. Under Risk Ratio scale

$$\begin{aligned} \text{NIE}_{\text{RR}} &= \frac{Q(1,1)}{Q(1,0)} = \frac{e^{\beta_2 \alpha_1 + 2\beta_2^2 \sigma^2}}{e^{\beta_2^2 \sigma^2}} = e^{\beta_2 \alpha_1 + \beta_2^2 \sigma^2} \\ \text{NDE}_{\text{RR}} &= \frac{Q(1,0)}{Q(0,0)} = \frac{e^{\beta_1 + \beta_2 \alpha_0}}{e^{\beta_2 \alpha_0}} = e^{\beta_1} \end{aligned}$$